Scaffold-controlled benchmarking of graph, transformer, and topological representations for antimalarial activity prediction

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Abstract

We tested whether graph, transformer, and topological molecular representations improve antimalarial activity prediction beyond an ECFP4–random-forest reference under chemical scaffold shift. We compared ECFP4–RF, a graph isomorphism network (GIN), GIN fused with topological fingerprints (GIN–TFP) or tensor-network embeddings (GIN–TNE), and a fine-tuned ChemBERTa transformer on 19 836 African antimalarial natural products using random and Bemis–Murcko scaffold five-fold cross-validation, with 25 fold–seed replicates per split. Under scaffold separation, ECFP4–RF achieved ROC AUC 0.830 0, exceeding GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7), and ChemBERTa (0.786 7); all four prespecified comparisons remained significant after Benjamini–Hochberg correction. Persistent-image dimensions showed the largest descriptive projection-weight salience in the topological fusion head, but this representation-level pattern did not improve ranking. A molecule-disjoint ChEMBL transfer panel reproduced the ECFP4–RF-over-GIN ordering within the broader antimalarial domain. These results provide a panel-specific, controlled negative benchmark: the conclusions apply to the evaluated data, splits, architectures, and training protocol, not universally to learned molecular models.

Contribution: This study quantifies the predictive and descriptive contributions of graph, sequence, and topological representations under scaffold separation, with ECFP4–RF serving as a reproducible reference. It also specifies fold-independent initialization for the pretrained transformer, a necessary condition for valid cross-validation.

Keywords: graph neural networks; molecular fingerprints; antimalarial natural products; scaffold split; benchmark; interpretability; persistent homology

Highlights:

- Under the scaffold split, ECFP4–RF (AUC 0.830 0) outperforms every GNN/transformer arm; all arms are significantly worse (BH-adjusted $p \leq 0.037\,8$)
- Fold-independent transformer initialization changes the apparent ranking of sequence models and is required for valid cross-validation

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- Persistent-image dimensions have the largest descriptive projection-weight salience in the topological fusion head
- Repeated evaluation across five seeds and five folds quantifies uncertainty under both split regimes

1 Introduction

Malaria remains a leading infectious disease burden in sub-Saharan Africa, with 247 million cases and 619 000 deaths in 2 023 [1]. Control is increasingly threatened by the emergence of artemisinin-resistant *Plasmodium falciparum* strains in Southeast Asia and Africa [2, 3], prompting an urgent, WHO-endorsed need for new scaffolds that remain active against drug-resistant parasites. African natural products (NPs) constitute a rich source of chemically diverse scaffolds [4, 5]. The panel studied here descends from a screening campaign seeded on 396 African natural products and 454 synthetic-drug scaffolds [6]. Computational screening of such libraries can accelerate hit identification, yet the choice of molecular representation directly conditions the apparent performance and chemical generalization of the predictor. In recent years, a "scale-centric" narrative has emerged in molecular machine learning: pretrained molecular foundation models and sequence transformers are expected to supersede compact, task-trained cheminformatics models [7, 8]. This narrative extends to quantum-inspired representations, where quantum computing is increasingly discussed as a paradigm for drug discovery and bioinformatics [9, 10].

Recent controlled benchmarks instead indicate that representation scale does not guarantee improved molecular prediction. In a 156-comparison benchmark across 26 endpoints, Guo and Ding found that classical machine learning on fingerprints wins 47.4 % of task–metric entries, ahead of pretrained sequence models (28.8 %) and GNNs (21.8 %), with the classical advantage largest under random splits and shrinking, but not vanishing, under scaffold-separated protocols [11]. An independent comparison of 25 pretrained embedding models across 25 datasets found that *almost no neural embedding statistically improves over the ECFP fingerprint baseline* [12]. And for natural products specifically, a 20-fingerprint benchmark over 100 000+ NPs showed that no single encoding dominates, and that ECFP is frequently matched or exceeded by alternative fingerprints on NP bioactivity tasks [13].

We therefore tested three linked hypotheses: whether learned models improve on ECFP4 for this chemically complex panel, whether persistent-homology and tensor-network descriptors reduce the scaffold-generalization gap when fused with GIN, and whether fold-independent initialization changes the apparent performance of a pretrained transformer.

1.1 Related work

The tension between classical cheminformatics and neural molecular learning has been documented across multiple benchmarks. For molecular property prediction, random forests on ECFP fingerprints remain competitive with or superior to GNNs on small-to-medium datasets ($< 10\,000$ molecules) [11, 12]. GNN architectures (GCN, GAT, GIN) have shown gains on large, curated datasets such as MoleculeNet [14], but these gains often diminish under scaffold splits that better reflect real-world drug discovery [11]. Pretrained sequence transformers (ChemBERTa, MolBERT) draw on large unlabeled corpora but have shown inconsistent gains on downstream tasks, particularly when the target dataset is structurally dissimilar to the pretraining data [12, 15].

Topological data analysis (TDA) has been applied to molecular representation through persistent homology, producing descriptors that capture ring systems, molecular cavities, and higher-order connectivity [16]. Recent work has fused TDA features with GNN architectures, reporting modest improvements on specific tasks [17]. Multimodal fusion of graph and sequence representations, for example concatenating GCN embeddings with ChemBERTa latent vectors, has shown promise on small datasets [18]. More broadly, computational MoA analyses emphasize that chemical, phenotypic, and network data capture complementary biological levels; target prediction alone is therefore not equivalent to a complete mechanism assignment [19]. Tensor-network embeddings, derived from Tucker decomposition of molecular tensors, provide compressed representations that retain scaffold-relevant information [8]. However, neither TDA nor tensor-network features have been systematically tested inside GNN fusion architectures under rigorous scaffold-split protocols on antimalarial panels.

We therefore benchmark GIN, GIN–TFP, GIN–TNE, and ChemBERTa against ECFP4–RF under both random and scaffold splits. We test three linked questions: whether learned models exceed ECFP4, whether topological fusion narrows the scaffold gap, and whether projection-weight salience identifies a reproducible geometric signal even when ranking does not improve. The comparison is designed to distinguish predictive performance from interpretability and to quantify the effect of fold-independent transformer initialization.

2 Results

2.1 ECFP4–RF defines the reference performance

ECFP4–RF provided the reference against which all learned representations were evaluated. Under the random split it achieved 0.9433 ± 0.0003 across five per-seed means (range 0.9428–0.9437), consistent with the previously reported performance for this molecular collection (0.9475 ± 0.0045) despite the distinct frozen split and random-seed realization [20]. This agreement supports comparability of the panel and fingerprint implementation.

2.2 An orthogonal phenotype-only MoA benchmark

We additionally evaluated a phenotype-only baseline on the public LISH mechanism-of-action (MoA) training release as an orthogonal analysis. This task is not a molecular representation benchmark: the release contains gene-expression, viability, dose, and time features but no versioned drug–SMILES mapping, precluding transfer claims for ECFP4, GNN, or transformer representations. After drug-level aggregation, the evaluation comprised 3 289 drugs and 206 scored MoA labels. An unweighted per-label logistic baseline was assessed with five seeds and five drug-grouped folds, using mean column-wise log loss as the primary metric and macro-AUPRC, macro-AUROC, Brier score, and expected calibration error as secondary metrics.

With vehicle controls retained, the phenotype-only baseline achieved mean column-wise log loss 0.02378, macro-AUROC 0.6435, macro-AUPRC 0.1428, mean Brier score 0.00374, and expected calibration error 0.00376 across 25 fold–seed evaluations. **These MoA-associated metrics are not numerically comparable to the P5 molecular ROC-AUC results because the task, labels, feature space, aggregation unit, and primary metric differ.** Removing vehicle controls produced nearly identical values (log loss 0.02389, macro-AUROC 0.6417, macro-AUPRC 0.1412); the change in macro-AUROC was only -0.0018. This sensitivity analysis supports the stability of the phenotype-only reference. Consistent with the distinction between phenotype, chemical, and network evidence in computational MoA analysis [19], these labels indicate MoA-associated prediction rather than causal target engagement.

2.3 ECFP4–RF outperforms learned representations under scaffold separation

Table 1 reports the mean test ROC AUC and the SD across five per-seed means for each model and split; each per-seed mean aggregates the five fold-level test AUCs. Under the scaffold split, the fingerprint baseline reaches 0.8300. Every graph-based and transformer arm is significantly *worse*: the best GNN arm, GIN–TFP, reaches 0.8138 ($\Delta = -0.016$, raw

Table 1: Mean test ROC AUC ± SD of the five per-seed means (each seed averages five fold-level test AUCs). The underlying analysis contains 25 fold–seed replicates per split. The displayed p-values are raw paired tests on the five per-seed means versus ECFP4–RF under scaffold; Benjamini–Hochberg correction across the four comparisons gives adjusted p-values of 0.0330 (GIN), 0.0330 (GIN–TFP), 0.0378 (GIN–TNE), and 0.0002 (ChemBERTa). All statistical comparisons are documented in the Methods.

Model	Random	Scaffold	Δ vs. ECFP4 (scaffold)
ECFP4–RF (baseline)	0.9433 ± 0.0003	**0.8300** ± 0.0023	—
GIN	0.9098 ± 0.0022	0.8047 ± 0.0141	−0.025 ($p = 0.018$*)
GIN–TFP (topological fusion)	0.9084 ± 0.0012	0.8138 ± 0.0107	−0.016 ($p = 0.025$*)
GIN–TNE (tensor fusion)	0.8918 ± 0.0018	0.8090 ± 0.0149	−0.021 ($p = 0.038$*)
ChemBERTa	0.9121 ± 0.0012	0.7867 ± 0.0054	−0.043 ($p < 0.0001$***)

paired $p = 0.025$; BH-adjusted $p = 0.0330$); GIN–TNE reaches 0.8090 ($\Delta = -0.021$, raw $p = 0.038$; BH-adjusted $p = 0.0378$); plain GIN reaches 0.8047 ($\Delta = -0.025$, raw $p = 0.018$; BH-adjusted $p = 0.0330$); ChemBERTa reaches 0.7867 ($\Delta = -0.043$, raw $p < 0.0001$; BH-adjusted $p = 0.0002$). All four scaffold comparisons survive Benjamini–Hochberg correction applied within that four-test family. Under the random split the same verdict holds: ChemBERTa reaches 0.9121 ($\Delta = -0.031$, raw $p < 0.0001$), GIN 0.9098 ($\Delta = -0.034$, raw $p < 0.0001$), GIN–TFP 0.9084 ($\Delta = -0.035$, raw $p < 0.0001$), and GIN–TNE 0.8918 ($\Delta = -0.052$, raw $p < 0.0001$), each significantly below ECFP4–RF (0.9433); all four random comparisons survive Benjamini–Hochberg correction applied within their own separate four-test family. *No graph-based arm outperforms the fingerprint baseline under either split.* A distribution-free check corroborates these parametric tests: paired permutation of the five per-seed mean AUCs against ECFP4–RF ($B = 10\,000$ permutations) returns a negative mean ΔAUC for every arm on both splits (two-sided $p = 0.009$ throughout), with bootstrap intervals that exclude zero — on the scaffold split, GIN −0.0253 (−0.0364–−0.0138), GIN–TFP −0.0162 (−0.0254–−0.0099), GIN–TNE −0.0210 (−0.0351–−0.0125), ChemBERTa −0.0433 (−0.0475–−0.0392) — computed from the per-seed arrays archived in the project's replication-statistics record.

The result was obtained with a fixed, repeated training protocol (AdamW, validation-based early stopping, selection of the validation-optimal epoch, and 25 fold–seed replicates). It is consistent with reports across ADMET/Tox21 endpoints [11] and pretrained-embedding benchmarks [12], in which local substructure encodings captured by ECFP4 and exploited by tree ensembles remain highly data-efficient.

2.4 ChemBERTa does not close the scaffold-generalization gap

ChemBERTa, fine-tuned with per-fold weight reset (see Methods), reaches 0.7867 ± 0.0054 under the scaffold split (SD across five per-seed means; five folds repeated over five seeds; fold-level range 0.724–0.836). Against the 0.8300 fingerprint baseline this is a significant deficit ($\Delta = -0.043$, paired $t(4) = -18.35$, $p < 0.0001$). Under the random split it reaches 0.9121 ± 0.0012, narrowly ahead of GIN (0.9098) but still 0.031 below ECFP4–RF (0.9433). The final scaffold hierarchy is thus:

$$\begin{aligned}\text{ECFP4–RF } (0.8300) &> \text{GIN–TFP } (0.8138) > \text{GIN–TNE } (0.8090) \\ &> \text{GIN } (0.8047) > \text{ChemBERTa } (0.7867).\end{aligned} \tag{1}$$

The ± values reported throughout are population standard deviations across per-seed mean AUCs under the fixed five-fold protocol; fold-level dispersions are correspondingly larger, and the raw per-fold statistics are archived in the replication table of the project record. These uncertainty summaries quantify repeated computational runs, not uncertainty over a future

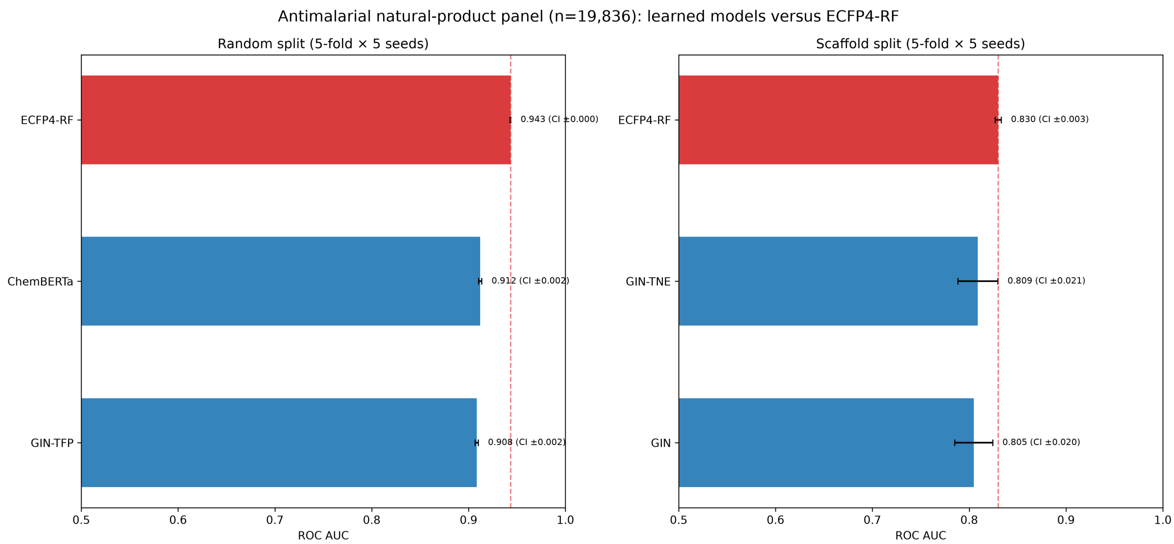

Figure 1: Mean test ROC-AUC with 95 % confidence intervals computed from five per-seed means (each mean aggregates five fold-level test AUCs), shown for random (left) and scaffold (right) splits. The underlying analysis contains 25 fold–seed replicates per model and split. Bar annotations report the mean followed by the CI half-width. The ECFP4–RF baseline (red) is not exceeded by any tested GNN or transformer arm.

population of compounds or datasets. The ordering among learned models is descriptive; the prespecified inferential comparisons were restricted to each learned arm versus ECFP4–RF.

Validation trajectories across 50 training epochs document the optimization path under the prespecified patience of 10; they describe optimization behavior rather than establishing the absence of over-fitting (Figure 2).

2.5 Topological descriptors remain visible without improving ranking

Even when predictive gains are absent, learned fusion weights carry structure that can be examined descriptively. We compute a *salience* score for each descriptor dimension as the mean absolute weight in the fusion projection layer (head_0) over the 25 replicates. For the TFP fusion (78 dimensions: 33 homology-dimension-0, 25 persistent-image, 20 Betti), the *persistent-image block (dims 33–57) has the largest raw mean projection-weight magnitude* (0.079 0) versus homology-0 (0.048 5) and Betti (0.054 7) blocks; the top five dimensions are persistent-image (42, 43, 52, 53, 54). For the TNE fusion (192 dimensions) the top dimensions are 68, 43, 92, 66, 165, and the top-10 % of dimensions account for 17.3 % of total salience (moderate sparsity).

This pattern indicates that topological descriptors are used by the fusion head, but it does not establish feature necessity or causal relevance. On this panel, the descriptor signal does not translate into a ranking advantage over fingerprints. Thus, projection-weight salience and predictive ranking should be treated as complementary, not interchangeable, summaries.

3 Discussion

Across the three hypotheses, learned representations did not exceed ECFP4 under either split; topological fusion narrowed the scaffold gap modestly but did not close it; and persistent-image dimensions remained salient in the fusion head without yielding a ranking advantage. Predictive utility and representation-level visibility are therefore separable properties. The separate LISH phenotype-only benchmark adds a complementary reference point: phenotype features support audited MoA-associated prediction, but the absence of a structure mapping prevents any claim

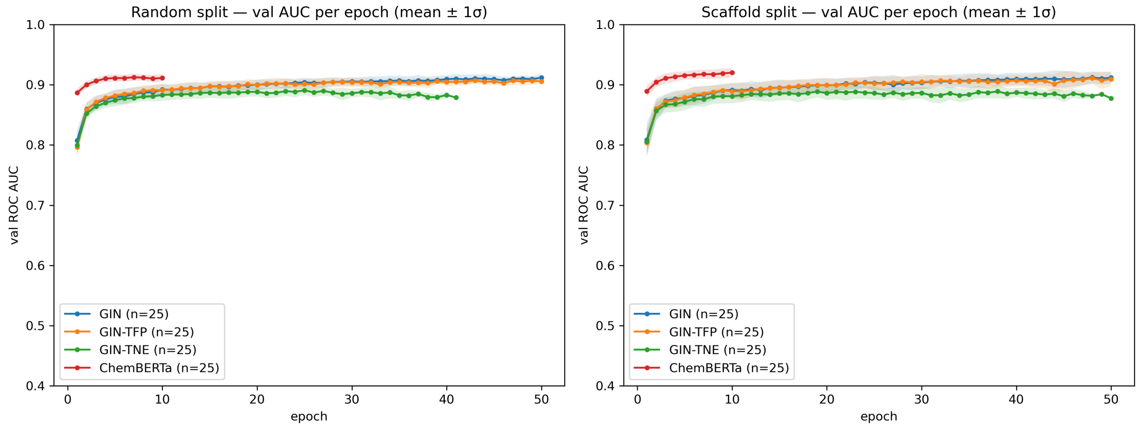

Figure 2: Validation ROC-AUC trajectories used to select the validation-optimal epoch. Curves are available for every tested arm under both random and scaffold splits; each available arm contains five folds repeated over five seeds. Early stopping with patience 10 selects the validation-optimal epoch before test evaluation.

that the molecular representations tested here improve MoA prediction. Its metrics are not numerically comparable to the molecular ROC-AUC results.

3.1 Performance depends on representation–task alignment

The primary result is that the tested pretrained and graph-based models did not improve ROC-AUC over ECFP4–RF on this panel, including under scaffold-held-out testing. This conclusion is supported by repeated evaluation, paired comparisons on the five per-seed means, and correction for multiplicity. The transformer result depends critically on fold-independent weight initialization: pretrained weights were reset at the beginning of every fold so that fine-tuning could not transfer between test folds. Future molecular-benchmark studies should state explicitly whether pretrained weights are reset for every fold.

3.2 Why fingerprints hold: local substructure dominance

The consistent ECFP4 advantage is best explained by representation–task alignment. ECFP4 encodes local substructure environments; for structurally related natural products, local SAR is the dominant signal, and tree ensembles exploit it efficiently with a few thousand labels [21]. Learned graph representations require more data to recover the same local patterns and generalize to scaffold-held-out series only partially [12]. This is consistent with the medicinal-chemistry view that fingerprints plus random forests remain an effective early-stage tool [22]. The counterpoint that graph models can be made competitive by training them *on* fingerprints is instructive: ECFP-pretrained GNNs improve out-of-distribution performance on homogeneous ADMET tasks but still underperform on heterogeneous endpoints such as binding affinity [23], mirroring the fingerprint-centric explanation, since pretraining transfers the very substructure knowledge fingerprints encode. The topological-fusion result adds a nuance: topological descriptors are not uninformative — they carry interpretable geometric signal, but their marginal ranking contribution is small on this panel.

3.3 Relation to prior antimalarial modelling

The molecular collection and fingerprint implementation were previously characterized in a quantum-representation benchmark [20]. That study found no statistically detectable advantage of the simulated quantum kernel over a tuned radial-basis kernel; the present analysis tests a

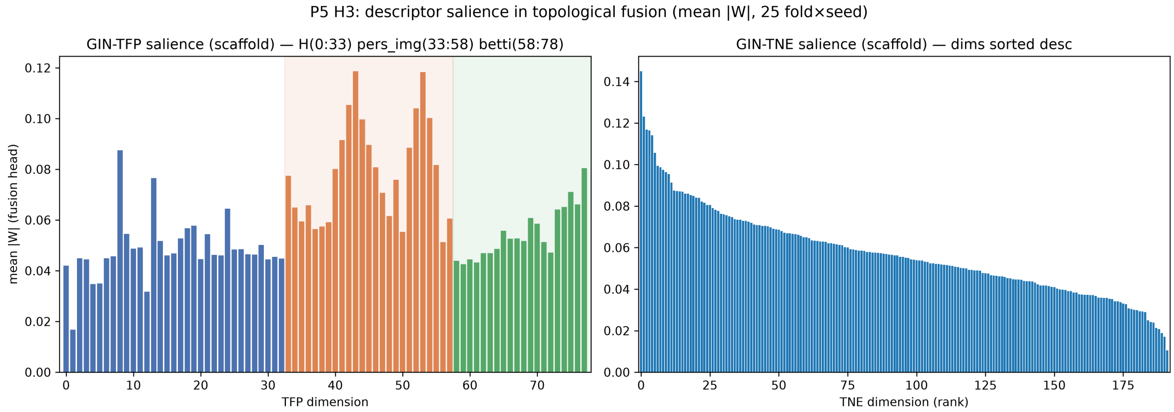

Figure 3: Descriptive projection-weight salience in the topological fusion heads under the scaffold split. Values are mean raw $|W|$ across 25 fold–seed replicates. Left: TFP (78 dimensions), partitioned into homology-0 (dims 0–32), persistent-image (dims 33–57), and Betti (dims 58–77); the persistent-image block has the largest raw weight magnitude. Right: TNE (192 dimensions), shown in descending order. Weight magnitude is interpreted descriptively and is not a causal attribution.

different question: whether graph or sequence models improve activity ranking on the same chemical collection. Molecular-dynamics studies of selected antimalarial leads have additionally examined resistance-associated targets and mutations [24]; those simulations provide biological context but are not labels or validation data for the present benchmark. The LISH result is complementary rather than directly comparable: it shows that a phenotype-only baseline yields stable MoA-associated predictions under drug-grouped evaluation, while the absence of molecular identifiers precludes testing whether ECFP4 or the learned representations transfer to that task. Topological descriptors are used descriptively by the fusion head, but none of the tested learned representations improves ROC-AUC over ECFP4–RF.

3.4 Implications for molecular-model selection

For comparable antimalarial NP panels, ECFP4–RF is a strong first-line baseline: it is computationally inexpensive and outperformed every tested neural arm. Graph and sequence models should be evaluated against this baseline under scaffold-held-out testing rather than assumed to improve from model scale alone. Transformer studies should reset pretrained weights for every fold and record the initialization protocol. Topological or tensor-network descriptors can be examined as complementary descriptive features, but their inclusion should not be interpreted as evidence of improved ranking without a paired comparison.

3.5 Limitations

Our panel, while curated and clinically motivated, is a single binary-activity collection assembled from screening data. To assess whether this ordering was specific to these computational labels, we repeated the ECFP4–RF versus GIN comparison on a disjoint ChEMBL-derived activity benchmark built from recorded IC_{50}/EC_{50} activities against *Plasmodium falciparum* (target CHEMBL364; compounds were labelled active at pChEMBL ≥ 6.0 and inactive at pChEMBL < 5.0, with the intervening 5.0–6.0 band discarded as ambiguous, retaining 22 447 of 31 649 unique compounds, and the 180 molecules sharing a canonical SMILES with the P5 panel were then removed, leaving 22 267 evaluation compounds). The fingerprint baseline again wins: ECFP4–RF 0.9190 ± 0.0004 versus GIN 0.8843 ± 0.0021 under the scaffold split ($\Delta = +0.035$, paired $p < 0.0001$). The fingerprint advantage is even larger on the external panel ($\Delta = 0.035$

versus $\Delta = 0.025$ on the canonical P5 panel), indicating that the ordering holds beyond the eOS80CH screening labels. This is a molecule-disjoint external transfer panel within the same broad antimalarial domain, not a cross-domain or prospective validation. Because this external analysis used an independently constructed split rather than the frozen internal fold files, it is a transfer analysis rather than a strict replication.

The GNN and fusion models use a single-layer prediction head and fixed hyperparameters (hidden dimension 128, 50 epochs, patience 10, and dropout $p = 0.1$ during training); alternative architectures or tuning could reduce the observed gap. Accordingly, the negative result applies to the evaluated configurations rather than to all possible GNN or transformer implementations. ChemBERTa was fine-tuned with a fixed learning schedule rather than per-task optimization. The GIN uses 2D molecular graphs without conformational information; 3D-equivariant architectures may perform differently on tasks for which shape is important [18]. The salience measure (mean absolute projection weight) captures correlation between descriptor dimensions and the predictive head, not causation; more sophisticated methods (e.g., integrated gradients, SHAP) would be needed to establish which features are truly necessary. Finally, the scaffold split is a proxy for novelty. The primary benchmark uses one frozen scaffold partition; a recent structural-frontier analysis shows that splits reserving the sparsest, physicochemically remote scaffold groups inflate error further on ADMET tasks [25]. As a versioned secondary robustness campaign, we generated three additional independent scaffold partitions and retrained the five GNN configurations (GIN, native and permuted GIN–TFP, and native and permuted GIN–TNE) on the canonical panel and on each new partition. Each configuration contained 25 fold–seed records with archived test probabilities, AUPRC, and individual salience vectors. Across the three new partitions, native GIN scaffold ROC AUC ranged from 0.788 6 to 0.796 7, native GIN–TFP from 0.801 3 to 0.810 5, and native GIN–TNE from 0.797 6 to 0.800 8; native AUPRC ranged from 0.902 5 to 0.916 0. These secondary estimates preserve the negative ordering relative to the canonical ECFP4–RF reference, but they are not replacements for the prespecified primary table. The separate extended ChemBERTa rerun was submitted after the GNN campaign and remains pending; no extended ChemBERTa metric is used here. Descriptor permutation reduced mean AUC numerically for GIN–TFP on all three new partitions (mean differences native minus permuted 0.005 4–0.013 9) and for GIN–TNE on two of three partitions (0.005 3–0.009 2); exact sign-flip tests on five per-seed means were not significant after accounting for the small computational unit. The extended campaign therefore supports, but does not prove, a modest contribution from the native descriptor arrays. Salience stability is now computable from the archived vectors: across the canonical scaffold and three new partitions, mean pairwise Spearman correlations were 0.755–0.829 for native TFP and 0.789–0.835 for native TNE, while mean top-10% Jaccard overlap was 0.374–0.399 and 0.274–0.326, respectively. These are stability diagnostics for weight patterns, not causal feature attributions. The standardization audit found no invalid SMILES or exact/parent duplicate rows in the 19 836-molecule panel, but identified 306 tautomer-collision rows; the primary labels were not changed. A versioned audit of the frozen splits found zero train–test scaffold overlap under the scaffold protocol, test-fold active prevalence ranging from 0.586 to 0.862, and bounded sampled mean maximum ECFP4 Tanimoto similarity of 0.292–0.388; these descriptive diagnostics were not used to alter the primary results. Two lightweight ECFP4 controls were also run on the same frozen splits. Distance-weighted kNN achieved scaffold ROC AUC 0.711 0 and AUPRC 0.846 7, while logistic regression achieved ROC AUC 0.706 3 and AUPRC 0.854 2; both ROC-AUC values were below ECFP4–RF, showing that the RF result should not be generalized to every ECFP4 learner. Under the random split, the corresponding kNN and logistic ROC AUC/AUPRC values were 0.916 6/0.956 2 and 0.879 0/0.945 6, respectively. These AUPRC values apply only to the lightweight controls and the separately versioned extended GNN campaign; the original canonical neural CSVs did not contain probabilities, so their AUPRC was recovered only through the new reruns and is not retroactively assigned to the historical canonical estimates. The separate extended ChemBERTa rerun was launched after the pretrained snapshot became available locally (SLURM array 15490); it remains pending and contributes no metric to this manuscript. A separate RF-only ChEMBL threshold audit was completed for four prespecified threshold specifications, with scaffold AUC ranging from 0.927 1 to 0.960 8; this is threshold sensitivity for the RF transfer analysis, not a full GIN threshold rerun. The LISH analysis is also intentionally limited: it is phenotype-only, uses aggregated MoA-associated labels, and has no drug–SMILES mapping. It therefore serves as an orthogonal pharmacological reference and control-sensitivity analysis, not as a molecular-model validation or evidence of causal mechanism.

4 Conclusions

In this scaffold-controlled benchmark of 19 836 African antimalarial natural products, ECFP4–RF outperformed the tested GIN, topological-fusion GNNs, and fine-tuned ChemBERTa transformer under both random and Bemis–Murcko scaffold cross-validation. Under scaffold separation, the hierarchy was ECFP4–RF (0.830 0) > GIN–TFP (0.813 8) > GIN–TNE (0.809 0) > GIN (0.804 7) > ChemBERTa (0.786 7); all four learned arms were significantly below the fingerprint baseline after multiplicity correction. The external ChEMBL-derived analysis showed the same ordering, while remaining an observational transfer analysis of thresholded activity records. Persistent-image dimensions had the largest projection-weight salience in the fusion head, indicating representation-level geometric structure without demonstrating feature necessity or improved ranking. Across the original aggregate salience summaries, the top 10

For comparable antimalarial natural-product panels, ECFP4–RF is therefore an appropriate reference model because it is inexpensive and retained the highest ranking performance in the present experiments. Claims of neural improvement should be tested under scaffold-held-out evaluation with paired uncertainty estimates. Topological and tensor-network descriptors may provide complementary descriptive information, but their inclusion should not be interpreted as a predictive gain without a direct paired comparison. Finally, pretrained transformer weights should be reset independently for every fold to prevent cross-fold information transfer. The separate LISH benchmark provides an audited phenotype-only MoA-associated reference, but its lack of molecular identifiers prevents a direct structure-to-MoA comparison; its metrics are not numerically comparable to the molecular ROC-AUC results. Experimental validation on prospectively synthesized candidates remains necessary to test whether the computational ranking predicts biological activity.

5 List of Abbreviations

ECFP4: extended-connectivity fingerprint, radius 2; **RF:** random forest; **GNN:** graph neural network; **GIN:** graph isomorphism network; **TFP:** topological fingerprint; **TNE:** tensor-network embedding; **MoA:** mechanism of action; **LISH:** public MoA benchmark; **ROC AUC:** receiver-operating-characteristic area under the curve; **CV:** cross-validation; **NP:** natural product; **ADMET:** absorption, distribution, metabolism, excretion, toxicity.

6 Methods

6.1 Data and panel

We use a frozen panel of 19 836 molecules with binary antimalarial activity labels from the eos80ch screening collection, curated and deduplicated by canonical SMILES as described previously [6]. Features are ECFP4 (2 048-bit Morgan radius-2), a 78-dimensional topological fingerprint (TFP; persistent homology, persistent images, and Betti descriptors), and 192-dimensional tensor-network embeddings (TNE). All features are aligned to the same molecule index.

Feature assembly fails rather than imputes. A SMILES string that RDKit cannot parse raises during graph construction; a non-finite value anywhere in the TNE matrix aborts the panel export; and a molecule absent from the precomputed TFP or TNE source table raises rather than receiving that column's mean. The third guard is the one that changes results. Mean imputation would leave the affected molecules carrying panel-average topology regardless of their actual structure while still contributing to the test metric, which inflates apparent performance for exactly the molecules whose descriptors are unknown. The panel is therefore defined to admit only molecules that carry their own measured features, and the released panel contains no imputed feature value. The external ChEMBL transfer panel is the single place this policy is relaxed: there an unparseable SMILES yields an empty graph so the benchmark completes over a public download we did not curate. No canonical result depends on that path.

6.2 Splits

Two protocols were evaluated, each with five folds repeated over five seeds (25 fold–seed replicates). (i) *Random*: stratified five-fold cross-validation. (ii) *Scaffold*: molecules were grouped by Bemis–Murcko scaffold before fold assignment. A Bemis–Murcko scaffold is what remains of a molecule once every side-chain substituent is deleted: its ring systems together with the linker atoms that join them [26]. Two molecules fall in the same group when they share that framework, however differently they are substituted. Scaffolds were computed with RDKit (`MurckoScaffold.MurckoScaffoldSmiles`), and whole scaffold groups were assigned to folds by greedy minimum-maximum size balancing, so the held-out test scaffolds are disjoint from the training scaffolds. Validation molecules were sampled from the non-test portion and therefore can share scaffolds with training molecules; early stopping consequently assesses optimization on the training chemical distribution, whereas the held-out test fold provides the scaffold-shift estimate. The split indices are released with the study.

6.3 Models

ECFP4–RF: the reference baseline, a scikit-learn `RandomForestClassifier` on the ECFP4 fingerprint matrix with 500 trees and `random_state=0`, fitted after `StandardScaler` inside the same pipeline, with scikit-learn defaults for every remaining argument including `class_weight=None`. **GIN**: graph isomorphism network (128 hidden units, 3 layers, ReLU activations, batch normalization after each layer, dropout $p = 0.1$ during training, mean-pooling readout, single head) trained on molecular graphs (atom/edge features from RDKit). We select GIN as the representative GNN architecture following MolGraphBench [11], which identified GIN as the optimal GNN for molecular regression across multiple benchmarks; GCN, GAT, and GraphSAGE were evaluated in that study and did not consistently outperform GIN, so we omit them to avoid redundancy. **GIN–TFP / GIN–TNE**: the same GIN with the topological descriptor vector concatenated at the head ($\text{head}_0 = \text{Linear}(128 + n_{desc} \to 128)$), enabling the salience attribution. **ChemBERTa**: pretrained SMILES transformer ('seyonec/ChemBERTa-zinc-base-v1'), fine-tuned with cross-entropy and independent pretrained-weight initialization for every fold. The GIN-family models used a maximum of 50 epochs with patience 10, batch size 512, learning rate 1×10^{-3}, and weight decay 1×10^{-4}. ChemBERTa used maximum sequence length 128, batch size 32, a maximum of 10 epochs, patience 3, learning rate 2×10^{-5}, and weight decay 0.01. All deep models ran on a single NVIDIA RTX A4000 GPU with AdamW and validation-AUC model selection.

6.4 Fold-independent transformer initialization

For each fold, the pretrained ChemBERTa state is restored before fine-tuning, so no fold receives weights updated on another fold. Without this reset, a fold's test molecules would already have been seen during an earlier fold's fine-tuning, and the reported test AUC would measure memorization as well as generalization. All transformer results use independent fold initialization and the same training schedule.

6.5 Orthogonal LISH MoA benchmark

We used the public LISH MoA training release as a separate, phenotype-only benchmark; the public challenge repository is available at https://github.com/LISHarvard/moa_challenge (accessed 12 August 2026). The preparation retained 772 gene-expression features, 100 cell-viability features, and 3 experimental-condition summary columns (exposure time, high-dose fraction, treatment-type fraction), and aggregated repeated assay observations to the drug level. The scored endpoint comprised 206 MoA labels for 3 289 drugs; labels were defined at the drug level from observed MoA-associated annotations. Because the public release used here did not contain a versioned drug–SMILES mapping, the benchmark was not used to train or evaluate molecular fingerprints, GNNs, or transformers. We fitted an unweighted logistic model independently for each label, evaluated with five random seeds and five drug-grouped folds, and reported mean column-wise log loss, macro-AUPRC, macro-AUROC, mean Brier score, and expected calibration error. A prespecified sensitivity analysis excluded vehicle-control samples, yielding 3 288 drugs. All fold records were audited for missing or non-finite metrics before interpretation. The endpoint is described as MoA-associated prediction because it does not establish causal target engagement [19].

6.6 Statistics

Per model and split, we retain the 25 fold–seed AUCs and summarize them as five per-seed means, each averaging the five fold-level test AUCs. Model-versus-baseline comparisons use paired t-tests on these five per-seed means ($df = 4$), rather than treating the 25 fold–seed values as independent observations. Table values are mean ± SD across the five per-seed means; the raw two-sided p-values are corrected with Benjamini–Hochberg separately across the four scaffold comparisons and the four random comparisons. As a distribution-free corroboration of these parametric comparisons, paired permutation tests were computed on the same five per-seed means (10 000 sign-flip permutations under a fixed seed); they are reported alongside the t-tests in Results. Under scaffold this yields adjusted p-values of 0.033 0 (GIN), 0.033 0 (GIN–TFP), 0.037 8 (GIN–TNE), and 0.000 2 (ChemBERTa); under random all four arms remain below the baseline after correction within their own separate family. The significance threshold is $\alpha = 0.05$ and effect sizes are reported as mean differences. Salience is aggregated across the 25 replicates as the per-dimension mean absolute projection weight.

Availability of data and materials

All code, frozen splits (random and scaffold, five seeds each), per-fold results, salience data, figure scripts, and the 19 836-molecule panel are available under the MIT license in the public GitHub repository (https://github.com/NanaEngo/Malaria_codesV2). The audited LISH phenotype-only fold outputs and comparison report are also released in the repository's P5 results artefacts. The official challenge source, derived-data provenance, and archive hash are recorded in the accompanying LISH README; the raw public archive and any credentials are not redistributed. Trained model states and feature tables are included; reproducibility is supported by the frozen split files and the pinned environment.

Secondary scaffold-only replication. A repeat scaffold evaluation of the ChemBERTa arm (five folds × five seeds, same code and same frozen splits as the primary run, distinct training run) produced all 25 fold–seed records and a mean ROC AUC of $0.790\,8 \pm 0.005\,8$ when summarized across five per-seed means; the primary scaffold estimate was $0.786\,7 \pm 0.005\,4$ under the same summary. The primary random-split estimate was 0.9121 ± 0.0012; no repeat random-split run was performed.

Author contributions

MVST conceived the study, designed the benchmark protocol, implemented the GNN/transformer pipelines, and wrote the manuscript. All authors contributed to data analysis, reviewed the results, and approved the final version. (Full author affiliations are given on the title page.)

Competing interests

The authors declare no competing interests.

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Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

Consent for publication

Not applicable. No individual person's data (including images, videos, or identifiable details) appear in this manuscript.

Use of Artificial Intelligence

The authors used AI-assisted tools during code development and data-analysis scripting. All affected text and code were reviewed and edited by the authors, who take full responsibility for the article. No AI system contributed as an author. The reported computations, statistical analyses, and quantitative claims were generated and verified by the authors.

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